

FAST – NUCES



Object Oriented Programming Lab Manual

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Lab 7 – Pointers to Class Members

Objectives

- Pointer to class object
 - Dynamic allocation of objects
 - Dot (.) vs Arrow (->) operator
 - Pointer to data member
 - Pointer to member function
-



Read carefully and attempt following task

Create a class named **Beyblade** inspired by the Beyblade cartoon.

Class Requirements

The class should contain the following **data members**:

- `int beybladeId` → represents the id number of Beyblade
- `float powerLevel` → represents the attack power of the Beyblade

Make these members **public** so they can be accessed using pointer-to-member concepts.

Task 1 – Basic Class and Object Pointer

Requirements

1. Create a class **Beyblade** with:
 - Default constructor
 - Overloaded constructor
 - `display()` function to print Beyblade information.
2. In `main()`:
 - Create an object of class **Beyblade**.
 - Create a **pointer of class type**.
 - Assign the pointer to the object.
 - Access the members using:
 - dot operator (`.`)
 - arrow operator (`->`)

Expected Concept

Students should understand:

`object.member`

`pointer->member`

Task 2 – Dynamic Allocation of Objects

Requirements

Inside main():

1. Dynamically create a **single Beyblade object** using new.
2. Initialize it using the **overloaded constructor**.
3. Call the display() function using **arrow operator**.

Example concept:

```
Beyblade* b1 = new Beyblade(101, 85.5);  
b1->display();
```

4. Delete the object using delete.

Task 3 – Dynamic Array of Beyblade Objects

Requirements

1. Ask the user for number of Beyblades.
2. Dynamically create an **array of Beyblade objects**.

Example concept:

```
Beyblade* list = new Beyblade[n];
```

3. Input values for each object.
4. Display all Beyblades using:

```
list[i].display();
```

5. Free memory using:

```
delete [] list;
```

Learning Goal

Array elements are objects, not pointers.

Correct usage:

```
list[i].display();
```

Incorrect usage:

```
list[i]->display(); // wrong
```

Task 4 – Pointer to Data Member

Requirements

Create a **pointer to data member** for powerLevel.

Syntax example:

```
float Beyblade::* ptr = &Beyblade::powerLevel;
```

In main():

1. Create a Beyblade object b1.
2. Use the pointer to modify powerLevel using:

```
b1.*ptr = 90.5;
```

3. Create a pointer to object:

```
Beyblade* bp = &b1;
```

4. Modify power level again using:

```
bp->*ptr = 120.0;
```

5. Display the object.

Task 5 – Pointer to Member Function

Create a pointer to the display() function.

Syntax

```
void (Beyblade::*fp)() = &Beyblade::display;
```

In main():

1. Create a Beyblade object.

2. Call the function using:

```
(b1.*fp)();
```

3. Also call it using pointer to object:

```
(bp->*fp)();
```

Important Notes for Students

1. **Dot Operator (.)**

Used with objects.

object.member

2. **Arrow Operator (->)**

Used with pointer to objects.

pointer->member

3. **Pointer to Data Member**

object.*pointer

pointer->*pointer

4. **Pointer to Member Function**

```
(object.*functionPointer)();
```

```
(pointer->*functionPointer)();
```

-----Happy Coding 😊-----



Submission Instructions

- Submit your tasks on Google Classroom within given deadline
- Submit solution file of each task separately for example “LAB01-TASK01.cpp”, “LAB01-TASK02.cpp” and so on
- “.cpp” is the actual file where all your code is saved.
- Always submit .cpp files when asked for submission
- Example path of .cpp file is
Desktop\Your_Roll_No\LAB01-TASK01\LAB01-TASK01\LAB01-TASK01.cpp
Copy your files from this path

